
GOMPERTZ GROWTH AS A SIGNATURE OF COHERENT, NON-LOCAL DYNAMICS IN BIOLOGICAL SYSTEMS

A PREPRINT

June 15, 2026

ABSTRACT

Logistic and Gompertz growth has traditionally been connected by augmenting the logistic model with an extra parameter to obtain what is known as θ -logistic growth or the Richards model for growth. In spite of this bridge, the biological foundation of Gompertz growth in the context of biological processes is only vaguely understood. To further our foundational understanding, we propose a novel microscopic version of the Richards model where the order parameter represents the coupling between nodes on a network. This model reveals Gompertz growth ($\theta \rightarrow 0$) as the unique coherent limit: the macroscopic dynamics decouple exactly from microscopic fluctuations, the system is unconditionally stable, and each entity contributes equally to the collective growth rate regardless of network topology. In contrast, Richards ($\theta > 0$) and logistic ($\theta = 1$) growth require synchronization as a precondition for macroscopic validity — a condition that Gompertz growth instead imposes. The coherence parameter θ thus plays the role of a symmetry-breaking parameter, with the Gompertz limit exhibiting infinite-order correlations in the natural domain, structural independence from microscopic noise, and a spectral gap protecting the macroscopic state — properties reminiscent of coherent states in quantum systems. These observations add to the interpretation of biological systems that undergo Gompertz growth where long-range interaction could be interpreted as the source of growth being external to the individual entities, like a field that stimulates growth simultaneously across all entities. This field could be interpreted as a stressor or stimulant, like an electromagnetic field in the short-term or time itself on a longer time horizon, acting on all entities simultaneously. Our new discovery allows for a unification of well-known growth models as follows: time independent and uncorrelated growth yields the exponential growth function, time dependent and uncorrelated growth yields Gaussian growth, time independent and pairwise correlated growth yields the logistic function and time independent and infinitely correlated growth yields the Gompertz function.

Keywords Gompertz · Coherence · Network Analysis · Self-organization

Highlights

- Logistic and Gompertz growth is connected within a novel framework based on microscopic principles.
- Gompertz growth is connected with a ubiquitous growth-activating field justified by physical and mathematical principles.
- The Gompertz model exhibits properties not exhibited by the logistic function, namely closure under aggregation, mean-variance decoupling, macroscopic stability, and egalitarian forcing from each microscopic entity.
- Growth processes are put in a larger contextual framework based on varying dependence in spacetime, including the Exponential, Gaussian distributions.

1 Introduction

In 1825, Gompertz published a paper on the nature of mortality where he discovered that from adulthood and onward “the power to avoid destruction” decreases exponentially with age. Equivalently, mortality rates increase in geometric progression for arithmetic increases in age, an observation now commonly referred to as Gompertz’ law of mortality [1]. Makeham (1889) later provided an interpretation to Gompertz’ law where the recuperative force of a human, or “vital force”, becomes less efficient with time as a “stressor” [2]. Gompertz’ law has later been known as the Gompertz model or Gompertz growth.

1.1 Gompertz’ law in existing work

This simple law has been shown to generalize to situations where mortality rates are substituted by many types of abundance variables, like the growth of animals, plants, bacteria, and cancer tumors [3–13], exposing Gompertz growth as nearly ubiquitous in biological growth processes. The idea that the Gompertz’ law can be applied to other biological systems was first proposed by Weymouth (1931) [3, 4] and Winsor (1932) [14]. Gompertz’ law has also been shown to apply to societal phenomena like growth in railway traffic, and the demand for goods and services, and sales of tobacco [15–17].

Alongside the many contexts wherein Gompertz’ law has been observed, an almost equal number of efforts have been made to interpret and derive this law from various phenomenological principles [18]. This effort has often been accompanied by a phenomenological comparison of another popular growth model called logistic growth, or Verhulst growth, where “the power to avoid destruction” decreases linearly with time, as opposed to decreasing exponentially with time as seen with the Gompertz model. In a study comparing the Gompertz and the logistic growth models, Petroni et al. (2020) [19] show that Gompertz growth can be interpreted as *maximally coherent* growth based on long-range interaction or correlation, while logistic growth is minimally coherent with non-collaborating elementary entities. With this interpretation, one is lead to postulate that the Gompertz model is a result of an external stimulus with perceived long-range interactions while the logistic model is a result of an internal stimulus with local interaction. As such, the logistic model has been popular both for communicable disease models [20–23] and for predator-prey theory [24], where disease or injury spreads through pairwise interactions.

A number of studies in the biological context have argued that Gompertz growth could be activated from an external factor which is ubiquitously and instantaneously introduced to the system, e.g. as a toxic agent or an environmental stressor to which the system gradually adapts [25–27]. In the original case of mortality rates as a function of age, this interpretation would imply that time itself is the environmental ubiquitous stressor. Other environmental stressors like chronic radiation exposure have been observed to shift the Gompertz curve, an observation also theoretically developed by Sacher (1956) ([28]). Sacher modeled the toxic stressor as a stochastic perturbation to the physiologic state of a population wherein death of an entity would occur beyond some perturbation threshold. The interpretation of Gompertz growth as the result of a ubiquitous stochastic perturbation was recently revived by De Lauro et al. (2014) [29] who arrived at the same result without alluding to a threshold value beyond which death was inevitable, but rather by letting the stochastic perturbation be the source of growth.

Another framework used to show how the Gompertz model can result from an external or ubiquitous stressor is chemical reaction theory. In reaction theory, a set of reactants combine to produce products, possibly in the presence of a catalyst. Markov (2019) [30] showed that the Gompertz model can be interpreted as growth under a catalyst whose effect diminishes at a rate independent of the growth rate of the reactant. In other words, the catalyst can be considered as an external and ubiquitous stimulant or stressor to the reactants, perfectly in line with the ubiquitous stochastic perturbation model from Sacher and De Lauro et al. On the other hand, if the logistic growth equation is recast into the reaction network equivalent, the catalyst is diminished by the growth of the reactant, which suggests that the source of growth is internal or at least dependent on the internal state. So also in reaction theory the Gompertz model could be seen as a result of a ubiquitous field while the logistic model could be seen as a result of internal interactions of the reactants.

It is worth noting that the notion that Gompertz growth is maximally coherent and non-local has also been derived from quantum mechanical principles [31]. Molski and Konarski (2003) showed that the Gompertz equation is the solution of a form of the time-independent Schrodinger equation. From this result, they derive the quantum coherent states which are non-local in space while moving along a classical time trajectory. This adds to our exhibits that the Gompertz model can be interpreted as non-local in origin. Coherent quantum states was originally formulated in the context of electromagnetic fields by Glauber [32] (for which he received the Nobel Prize). Some biological macroscopic growth phenomena are also due to such fields. For example, the role of internal electrical fields in regenerative growth is well established and is based on negative electrical potentials that act on sensitive cells [33]. In fact, it has been shown that cancer malignancies are mostly *negative* in electric polarity with respect to the rest of the body, while nonmalignant (or benign) tumor pathology is mostly *positive* in polarity [34]. Gompertz growth has also been widely observed

in cancerous cells [35, 5]. One is then led to the interpretation that certain biological growth processes that exhibit Gompertz growth possibly are coupled with an electromagnetic field of specific character, or e.g. a collective nutrient deprivation. Consequently and unsurprisingly, the artificial disruption of such a field has been found to disrupt the mitotic cell cycle [36–39]. This theory would also validate earlier cell-based theories where Gompertz growth in tumors are postulated to arise from an inverse relationship between diversity in cell states and the number of intercellular interactions [40]. In other words, where a maximally interacting system leads to a minimum of possible states, or vice versa, the Gompertz emerges.

On the topic of disease modelling, the Gompertz model has been shown by Wang et al. [41] to be incompatible with a subset of models in the communicable disease paradigm. They used the common Susceptible-Infected-Recovered (SIR) model by Kermack and McKendrick [42] augmented by the Richards parameter [43] and showed that the allowed parameter range within this model does not cover the parameter value taken in the Gompertz scenario. Other work incorporates network heterogeneity, super-spreading, and behavioral feedback to model sub-exponential/Gompertz-like dynamics in transmission models [44–46]. If one incorporates a social proximity network on which the disease spreads, Zonta and Levitt [47] observed through computer simulations that a near-Gompertz growth can occur after an initial transient period but only for scale-free networks, a type of network that some claim to be rarely representative of social networks [48–50] and others claim to be common [51], with varying degrees of rigor. We will show that one can obtain an exact Gompertz curve from a ubiquitous stressor scenario based on microscopic interactions regardless of network topology. While many studies show the SIR model (and its extensions) as the fitted model in epidemics [52–57], some recent studies have successfully fitted a Gompertz model [58–62].

1.2 Main aim

In this paper, our primary aim is to investigate the source of growth in self-organized processes and develop a microscopic version of the Richards model. In particular, we posit that the source of growth can be elucidated by considering interactions in the microscopic domain which can be aggregated to recover the traditional macroscopic growth equations. Second, we will elaborate on the idea that the logistic model results from local internal interactions while the Gompertz model results from a possibly external field or equivalently from an infinite set of higher order interactions between the microscopic entities involved in the growth. We will extend the Richards parametrization of the Gompertz-Logistic equations to apply in the microscopic domain with interacting entities, building on previous work in the macroscopic domain [19, 63, 41]. In presenting this work, we present a novel bridge between the logistic and the Gompertz model which offers both a mathematical and biological foundation to the source of growth.

1.3 Roadmap

Our paper is structured as follows: we will first introduce the macroscopic Gompertz-Logistic models with the traditional Richards parameter as the bridge between them. We then present a microscopic extension adding a local coupling term between microscopic neighbors. The interpretation of such a coupling depends on which model we use. In the logistic extreme, the coupling term represents local/pairwise transmission propagation, as needed in e.g. contagion dynamics, while in a Gompertz extreme the same term represents the shared response to a common exposure where every entity is pulled toward the collective state equally regardless of which is ahead or behind, because the stressor acts on every entity simultaneously rather than propagating from entity to entity.

We further show that only the Gompertz model is 1) closed under microscopic aggregation and 2) results from non-local simultaneous long-range interactions, which could alternatively be seen as a result of ubiquitous field or in the biological context: an environmental stressor, internal or external. Then, we explore mathematical properties exhibited by the Gompertz model and prohibited by the logistic model, like stability in the face of internal fluctuations, mean-variance decoupling, both while introducing a microscopic interaction term in the model. We proceed with a discussion that summarizes our findings in the context of the other well-known growth modalities involving the Gaussian and the Laplacian model.

2 Theory: From Macroscopic to Microscopic Richards growth

2.1 Macroscopic model

First consider a system that grows according to logistic growth. If we let the variable $X(t)$ denote the abundance of a quantity in a normalized system where its maximum size is 1, the equation that determines its size as a function of time is

$$\frac{1}{X} \frac{d}{dt} X(t) = \beta(1 - X(t)), \quad (1)$$

where $\beta > 0$ is a growth parameter. Here, the right hand side is commonly interpreted as representing “resources available for growth” which decrease linearly with respect to the relative abundance’s growth rate on the left hand side. One way to generalize logistic growth is to introduce a parameter that exponentiates the resource availability in the following way,

$$\frac{1}{X} \frac{d}{dt} X(t) = \frac{\beta}{\theta} (1 - X^\theta(t)), \quad (2)$$

also called θ -logistic growth or Richards model [43]. Petroni et al. (2020) argued that θ can be interpreted as a parameter capturing non-linear interaction effects. They showed that in the limit where the system becomes *maximally cooperative* or *coherent*, $\theta \rightarrow 0$ and the θ -logistic equation reduces in this limit to the Gompertz growth model,

$$\frac{1}{X} \frac{d}{dt} X(t) = \frac{d}{dt} \ln X(t) = -\beta \ln X(t). \quad (3)$$

In this limit, the relative growth rate goes to infinity as $\lim_{X(t) \rightarrow 0}$, which could be seen as a verification of the maximally coherent characteristic of the system. One is reminded of a system that is perturbed and left to reach a new equilibrium. Petroni’s definition of maximal coherence in the context of the Gompertz equation was shown to rigorously correspond to the definition of coherence used in quantum mechanical systems [31] attributed to Glauber (1963) [32].

2.2 Microscopic model

Given that both the logistic and Gompertz models describe a sigmoidal saturation at the macroscopic level, what distinguishes them at the microscopic level? The answer will turn out to be the domain in which dynamics are linear, which in turn determines the correlation structure. First, note that the Gompertz function is naturally expressed as a polynomial in the log-domain $\ln X(t)$, while the logistic function is naturally expressed as a polynomial in the original domain. In the logistic model in Eq. 1, arithmetic growth, $dX(t)/dt$, is effectively driven by the interaction between the two macroscopic states, $X(t)$ and $1 - X(t)$, also known as the transmission term in an SIR model.

Let us split the system into N microscopic abundance variables, $x_1(t), x_2(t), \dots, x_N(t)$ (sometimes we omit t for brevity), which are unitless and take values between 0 and 1, where 0 represents no realized growth while 1 represents the maximum potential growth having been realized. In other words, these variables represent the *realized potential*, either in the form of individual volume, mass, likelihood of infection or mortality by being preyed on, etc.

A key point is that only Gompertz growth is *closed* under the geometric mean of the microscopic equations, i.e. both micro- and macroscopic domains obey the same dynamics. To see this, note that macroscopic Gompertzian growth is retained when we compute the arithmetic mean in the log-domain, with $z_i = \ln x_i$,

$$\ln X(t) = \frac{1}{N} \sum_i^N z_i(t) \quad (4)$$

$$\frac{d}{dt} z_i(t) = -\beta z_i(t) \quad (5)$$

$$\therefore \frac{d}{dt} \ln X(t) = \frac{1}{N} \sum_i \frac{d}{dt} z_i(t) = -\beta \frac{1}{N} \sum_i z_i(t) \quad (6)$$

$$\frac{d}{dt} \frac{1}{N} \sum_i z_i(t) = -\beta \ln X(t). \quad (7)$$

The same closure does not hold for logistic dynamics. To see this, we must apply the microscopic logistic model with communicates between x_j and x_i via a coupling constant w_{ij} that is assumed both row- and column-normalized $\sum_i w_{ij} = \sum_j w_{ij} = 1$, i.e. the total influence received from node j and returned out to the network across all nodes sums to 1,

$$\dot{x}_i = \beta(1 - x_i) \sum_j w_{ij} x_j. \quad (8)$$

We call this coupling matrix doubly-stochastic [64], and it can be interpreted as a regular network, where each node shares the degree distribution, as opposed to a non-regular network where nodes have different degree distributions and strengths. Non-regular networks would not be normalizable in both dimensions simultaneously.

Notwithstanding network regularity, the coupling mechanism is absolutely necessary in a transmission model, but not in an external stressor model where a ubiquitous field can be regarded as the catalyst or trigger. So while we must include coupling with a network in a logistic model involving microscopic dynamics it is not necessary in Gompertzian

dynamics. However, we shall see in the sequel that even when we include network coupling in the Gompertz model closure still holds for some networks.

Now, we proceed with the logistic model to average over the microscopic entities, with $\bar{x} = \frac{1}{N} \sum_i x_i$,

$$\frac{1}{N} \sum_i \dot{x}_i = \frac{\beta}{N} \sum_i (1 - x_i) \sum_j w_{ij} x_j = \frac{\beta}{N} \sum_i \sum_j w_{ij} x_j - \frac{\beta}{N} \sum_i \sum_j w_{ij} x_i x_j. \quad (9)$$

The first term is simply $\frac{\beta}{N} \sum_j x_j = \beta \bar{x}$.

Now, for closure to hold, we need the right hand side to equal $\beta \bar{x}(1 - \bar{x})$. The second term is the weighted average of pairwise products, $\frac{\beta}{N} \sum_i \sum_j w_{ij} x_i x_j$, which is $\beta \bar{x}^2$ only if either x_i and x_j are uncorrelated across the network, $\text{Cov}(x_i, x_j) = 0$, or all nodes are completely synchronized i.e. $x_i = \bar{x}$ for all i . Note that non-regular networks also exhibit this lack of closure in the logistic setting.

Neither synchronicity or lack of correlation is compatible with most scenarios the microscopic logistic model is meant for, e.g. predator-prey dynamics or communicable diseases. Synchronization precludes any heterogeneity in abundance or health status, while the uncorrelated condition requires that abundance be uncorrelated with its neighbors — contradicting the very transmission mechanism on which communicable disease models are predicated and similarly for predator-prey models. In a real epidemic, highly connected nodes are infected earlier, creating strong positive network-state correlations that violate both conditions. Previous work has shown near-Gompertz growth with a scale-free network in computer simulations [47], but theoretical equivalence has yet to be shown and transient non-Gompertz behavior is seen initially in those simulations. Scale-free networks are also rare in social networks [48–50].

Instead, the proposed microscopic Gompertz model implies higher-order dependence between microscopic variables without any new parameters. This is seen by examining the partial derivatives of the geometric mean with respect to the abundance variables,

$$X(t) = \left[\prod_i x_i(t) \right]^{1/N} \quad (10)$$

$$\therefore \frac{\partial}{\partial x_i} X(t) = \frac{1}{N} \frac{X}{x_i}, \quad (11)$$

and so any subsequent partial derivative with respect to another $x_j \neq x_i$ will also be a function of all $(x_1, x_2, \dots)$, with no order at which the dependence truncates. This is a mathematical property of the geometric mean as an aggregation operator: the log-normal distribution in the abundance domain has nonzero cumulants at all orders, unlike the Gaussian distribution that arises from arithmetic averaging. In contrast, the arithmetic mean has $\partial X / \partial x_i = 1/N$ with all higher mixed partials equal to zero, so the logistic model's macroscopic observable couples to microscopic states only at first order.

We emphasize that this infinite-order dependence is a necessary consequence of the dynamics being linear in the log-domain. The microscopic Gompertz dynamics in $z_i = \ln x_i$ are linear and only pairwise-coupled, producing a jointly Gaussian distribution in z -space with correlations fully characterized at second order. The exponential map $x_i = e^{z_i}$ then transforms this second-order structure into the infinite-order cumulant structure of the log-normal distribution in the original domain. Thus, "infinite correlations" are generated by the coordinate transformation between the natural dynamical domain (z -space) and the observable domain (x -space), not by explicit higher-order interaction terms in the equations of motion.

For the microscopic Gompertz model, the aggregation operator is determined by the condition that the dynamics should be closed under aggregation, thus the geometric mean is not a modeler's choice but rather an operation that preserves dynamics. That logistic growth is not exact at the macroscopic level implies some noise is expected in real observations or simulations (see Results section).

In brief, the emergence of the Gompertz curve at the macroscopic or collective level suggests that the system is correlated, or coherent, presumably as a result of the simultaneous exposure to the same underlying stressor and due to the implied log-normal nature of the microscopic entities, where multiplication replaces addition as the aggregating operator. Thus, infinitely correlated growth is a natural consequence of macroscopic Gompertzian dynamics, which we here choose to call *coherent growth* in accordance with previous work [65, 66, 31]. We have thus showed *what* distinguishes the logistic from the Gompertz model, namely the closure conditions and the correlation order, and now turn to show *how* these models connect through the continuous Richards parameter.

3 Unifying microscopic logistic growth with Gompertzian growth

The Richards model unifies the macroscopic logistic and Gompertzian growth models, but by proposing a microscopic version of this model we shall see that Gompertz growth at the microscopic level is the only type of growth which is coherent in the sense that a) internal fluctuations do not perturb the macroscopic mean, b) the variance of the fluctuations is decoupled from the macroscopic mean, and c) each entity contributes equally to the global growth of the system. Let us start with N nodes in the log-transformed domain, $z_i = \ln x_i$, connected by a network of edges represented by our doubly stochastic connectivity matrix, W , where the ij^{th} element is the positive coupling strength, between node i and j . With these assumptions, the following microscopic model for growth is postulated,

$$dz_i = \left[\frac{\beta}{\theta} (1 - e^{\theta z_i}) + \beta(\bar{z}_{\theta,i} - z_i) \right] dt + \sqrt{\sigma} dB_i(t) \quad (12)$$

where $\bar{z}_{\theta,i} = \frac{1}{\theta} \ln(\sum_j w_{ij} e^{\theta z_j})$ is the order- θ power mean of neighbors' states. The parameter θ smoothly interpolates between Gompertz ($\theta \rightarrow 0$) and logistic ($\theta = 1$) dynamics, with intermediate values recovering the Richards growth family. The coupling is designed so that θ -dependence is consistent with the macroscopic Richards interpolation. The noise, $dB_i(t)$ is assumed to be independent normally distributed, interpreted as an external stressor having a unique response on each individual modulated by noise strength σ . This noise structure is multiplicative as opposed to additive and thus retain positive abundance values. Note that for perfectly synchronized nodes, with $z_i = \bar{z}$, Equation 12 reduces to a single-entity model, as seen in De Lauro et al. [29], wherein the noise term is interpreted as the source of growth itself. If $z_i \neq \bar{z}$, then there are two sources of growth, one stochastic and one through network effects.

First, we see that in the unique case of Gompertz, i.e. $\theta \rightarrow 0$, the model is still closed under aggregation even with the coupling term. This is because for the doubly stochastic property of the adjacency matrix, the second term goes to zero because $\frac{1}{N} \sum_i \sum_j w_{ij} z_j = \bar{z}$ and the third term goes to zero by the law of large numbers.

Secondly, any microscopic fluctuations revert to the mean, producing a stable evolution independent on the fluctuation profile. We see this by first defining a set of zero-mean fluctuations to be

$$\delta_i = z_i - \bar{z} \quad (13)$$

then,

$$\frac{d\delta_i}{dt} = \frac{dz_i}{dt} - \frac{d\bar{z}}{dt} \quad (14)$$

$$= -2\beta(\bar{z} + \delta_i) + \beta \sum_j w_{ij}(\bar{z} + \delta_j) + \beta\bar{z} \quad (15)$$

$$= -2\beta\bar{z} - 2\beta\delta_i + \beta\bar{z} \sum_j w_{ij} + \beta \sum_j w_{ij}\delta_j + \beta\bar{z} \quad (16)$$

$$= -2\beta\delta_i + \beta \sum_j w_{ij}\delta_j, \quad (17)$$

where we have used the assumption that columns of w_{ij} sum to 1. The result is a linear system $\dot{\delta} = \beta(-2I + W)\delta$. The eigenvalues of W for a row-normalized matrix lie in $[-1, 1]$, so the eigenvalues of $-2I + W$ lie in $[-3, -1]$. All eigenvalues are strictly negative, meaning every deviation mode decays exponentially with rate at least β . Thus, one could say the system is unconditionally stable, where deviations shrink regardless of their initial magnitude.

Combining the first and the second point we see that the Gompertz model does not require synchronicity, but rather imposes it. One could call this a characteristic of a coherent system.

Thirdly, the Gompertz model's mean and variance are decoupled, which is crucially not the case for any other model in the Richard spectrum where $\theta > 0$. To see this, first recognize that the variance of z_i is $\sum_j w_{ij} \delta_j^2$, our zero-mean random fluctuations. Then, by Taylor expanding the order- θ mean around $\theta = 0$, and using the property that δ_i has mean of 0,

$$\bar{z}_{\theta,i} = \frac{1}{\theta} \ln \left(\sum_j w_{ij} e^{\theta z_j} \right) = \frac{1}{\theta} \ln \left(e^{\theta \bar{z}} \sum_j w_{ij} e^{\theta \delta_j} \right) = \bar{z} + \frac{1}{\theta} \ln \left(\sum_j w_{ij} (1 + \theta \delta_j + \theta^2 \delta_j^2 / 2 + O(\theta^3)) \right) \quad (18)$$

$$\bar{z}_{\theta,i} = \bar{z} + \frac{\theta}{2} \text{Var}(z_i) + O(\theta^2), \quad (19)$$

and we immediately see that the mean and the variance are uncoupled only when θ goes to 0. Thus, for the general- θ Richard's model, only if all variables are in complete synchronicity, $z_i = \bar{z}$, will the microscopic model reduce to the macroscopic model, and if $\theta = 0$, a single entity Gompertz model. Overall, the macroscopic Richards model is *conditioned* on synchronicity, while the Gompertz model *imposes* synchronicity in the deterministic case ($\sigma = 0$) and maintains exact macroscopic reduction despite residual fluctuations in the stochastic case, with $\text{Var}(\bar{z}) = \frac{\sigma}{4\beta N}$. This is due to each δ_i having variance $\sigma/4\beta$ and being weakly correlated for large N , so that when the variance of the mean is computed the answer is as shown.

Lastly, we show that each entity equidistant from the mean contributes equally to the growth in the Gompertz model, an "egalitarian" property. We first present a simple example. Consider a two-variable system at a particular point in time where they are at unequal abundance, e.g. $x_1 = 0.01$ and $x_2 = 0.1$. In this state, $\bar{z} = (\ln 0.01 + \ln 0.1)/2 \approx -3.45$ which, upon computation of the other terms, reveals equidistance in additive terms in the log-domain and in multiplicative terms in the original domain; $z_1 - \bar{z} = \bar{z} - z_2$ and $x_1/e^{\bar{z}} = e^{\bar{z}}/x_2$. This implies that the second term in Eq. 12, the collective forcing term, has the same magnitude for both variables. The same could not be said in the logistic growth model, where the mean would be the arithmetic mean of the two states, 0.055, and closer to the higher value.

To show this generally, we just have to show that this collective growth term, $\beta(\bar{z}_{\theta \rightarrow 0, i} - z_i)$ is independent of $z_i \forall i$ and only a function of the fluctuations, δ_i . We use the fluctuation from the mean $z_i = \bar{z} + \delta_i$, viz.

$$\begin{aligned} \bar{z}_{\theta \rightarrow 0, i} - z_i &= \lim_{\theta \rightarrow 0} \frac{1}{\theta} \sum_j w_{ij} e^{\theta z_j} - z_i \\ &= \sum_j w_{ij} z_j - z_i \\ &= \sum_j w_{ij} (\bar{z} + \delta_j) - (\bar{z} + \delta_i) \\ &= \sum_j w_{ij} \delta_j - \delta_i \end{aligned}$$

where the first right-hand-side term vanishes in expectation in the large- N limit, thus reducing the collective growth term to

$$\lim_{N \rightarrow \infty, \theta \rightarrow 0} \bar{z}_{\theta \rightarrow 0, i} - z_i = -\delta_i, \quad (20)$$

which completes the argument.

In contrast, logistic growth ($\theta = 1$) has a coupling term $\bar{z}_{1, i} = \ln(\sum_j w_{ij} e^{z_j})$ which depends on all neighbours and any node $z_i \gg \bar{z}$ will dominate the exponential sum. Thus, in logistic growth the outliers drive the growth of the system, while in Gompertz growth, growth is egalitarian, and the forcing magnitude is the same for nodes equidistant from the mean regardless of which specific neighbors they have.

3.1 Dropping the assumption of double stochasticity

Without the doubly stochastic network assumption both the mean-variance decoupling and the mean reversion of the fluctuations still hold as they require only row-normalization. Only the first property, closure under aggregation, will retain the extra term $\frac{1}{N} \sum_i \beta(\bar{z}_{\theta, i} - z_i)$, but due to the synchronicity property for the Gompertz model, this term will go to zero as time progresses. The rate of this decay, or the closure discrepancy, depends on network heterogeneity in the following way. Letting k_i be the number of neighbors for node i , $\mathcal{N}(i)$ be the set of those nodes, and $c_j = \sum_{i \in \mathcal{N}(j)} 1/k_i$ be the network weight for this node,

$$\epsilon_G \approx \left| \frac{1}{N} \sum_i \beta(\bar{z}_{\theta, i} - z_i) \right| = \left| \frac{1}{N} \sum_j (c_j - 1) z_j \right|, \quad (21)$$

where the random error terms $dB_i(t)$ average to zero approximately (hence the approximate-equality sign). Note that for a regular homogeneous network $c_j = 1 \forall j$. This, we can express the discrepancy approximately as a function of the covariance between the degree heterogeneity and the node states,

$$\epsilon_G \approx \beta \left| \frac{1}{N} \sum_i c_i z_i - \bar{c} \bar{z} \right| = \beta |\text{Cov}(\mathbf{c}, \mathbf{z})|, \quad (22)$$

with a slight abuse of notation as these are not random vectors, but rather statistical properties.

We now see that both large network heterogeneity or large node variance results in large deviations from the macroscopic model. For example, consider a network where there are many nodes with low degrees and few neighboring nodes in a local coupling. Such low-degree nodes will synchronize tightly amongst themselves but not towards the global mean, thus creating a higher covariance and closure discrepancy. However, simulation results show that the mean-field approximation is good even for quite heterogeneous, non-regular real networks.

The corresponding closure discrepancy for logistic growth depends directly on the node states' cross-covariance, shown from Eq. 9,

$$\epsilon_L \approx \beta \left| \frac{1}{N} \sum_j \sum_i w_{ij} x_i x_j - \bar{x}^2 \right| = \beta |\text{Cov}_W(\mathbf{x}, \mathbf{x})|, \quad (23)$$

where the covariance includes the network weights for each node. The logistic model's suspected larger closure discrepancy arises from three reinforcing mechanisms — mean-variance coupling that prevents the macroscopic equation from closing, absence of the self-correcting synchronization that the Gompertz model uses to eliminate network-dependent corrections, and asymmetric forcing that will be exacerbated when there is a large momentum and thus microscopic variance in the growth process. These three effects are absent in the Gompertz limit, which is why the closure discrepancy is both smaller and self-correcting.

In brief, neither Gompertz or logistic close without a mean-field approximation, i.e. for non-regular networks, although the logistic model is suspected of having a larger closure error and thus would exhibit more instability than the Gompertz. With regular networks, the Gompertz exhibits closure while the logistic model still enjoys no such property. Further elaborations on rate of decay of closure discrepancy is left for future work.

4 Unifying the Gaussian and Exponential models with the Logistic and the Gompertz models

Up until now we have seen how logistic growth emerges from pairwise correlated growth while Gompertz growth emerges from infinitely correlated growth. These are both growth modes where there is some degree of connectivity between the entities. But what if there is no connectivity between the entities but each entity is independent? In such a case we can formulate a growth equation where the relative growth is only linearly dependent on time, where we make a slight modification and model the decay or *potential abundance*, $s(t) = 1 - x(t)$, such that $s(0) = 1$,

$$\frac{d}{dt} s(t) = -\beta t s(t), \quad (24)$$

or

$$\frac{d}{dt} \ln s(t) = -\beta t, \quad (25)$$

for some constant $\beta > 0$.

Solving this equation yields the familiar Gaussian function,

$$s(t) = A e^{-\beta t^2/2}, \quad (26)$$

where β can be interpreted as the inverse variance or dispersion of the growth phenomena and where $A = 1$ when $s(0) = 1$. Thus, not surprisingly, Gaussian growth emerges for mutually independent events that have a linear time dependence. The same result was originally formulated by Gauss with space as the independent variable instead of time and with spatial estimation error as the dependent variable instead of abundance [67] (see [68] for details). Another example of space as the dependent variable in the Gompertz setting is given by Molski and Konarski [31].

If we remove the dependency on time altogether, we observe the exponential growth equation

$$\frac{d}{dt} \ln s(t) = -\beta, \quad (27)$$

which yields

$$s(t) = A e^{-\beta t}. \quad (28)$$

This exponential equation implies a complete independence on both time and of the entities involved in the growth. In the context of measurement error this function would represent the Laplacian error which states that absolute measurement error decreases linearly with the error itself¹.

¹Or as Laplace originally stated: “[...] we have no reason to suppose a different law for the ordinates than for their differences” [68].

To summarize we have

$$\frac{d}{dt} \ln s(t) = -\beta \cdot g(s, t) \quad (29)$$

where g encodes growth rate dependency, or the source of growth:

- Exponential: $g = 1$ (nothing)
- Gaussian: $g = t$ (time only)
- Logistic: $g = 1 - s$ (abundance only, linear)
- Gompertz: $g = -\ln s$ (abundance only, logarithmic)

Note here that closure under aggregation is only relevant where there is inter-entity coupling, and since neither the Exponential or Gaussian scenarios have such coupling, closure occurs trivially, and microscopic treatment thereof becomes superfluous.

Simulations and Results

Simulated microscopic growth scenarios for a variety of Richards "coherence" θ parameters all show close correspondence to their theoretical counterparts, except at high values of θ where the mean-variance coupling identified in Eq. 19 produces visible deviations from theory, particularly in the late stages where the double-log transform amplifies residual fluctuations near saturation (Figure 1). The numerical component of this deviation can be mitigated with adaptive sub-stepping, but the physical component — arising from the mean-variance coupling — persists and is left visible to illustrate this structural difference between the Gompertz and logistic regimes. We do not see a large difference when using different networks in the simulations, consistent with the mean-field approximation described in the Methods section, under which the network topology enters only through degree-weighted averages (Figure 2).

Comparing the mean-field approximation with a non-approximating realistic network simulation does not yield notable discrepancies (Figure 3), further validating the approximation scheme. The explicit and mean-field results are indistinguishable across all topologies, including the Scale-Free network (max degree 417) and Power Law network (max degree 14,675), confirming that the macroscopic Gompertz dynamics are robust to actual network structure and do not depend on the doubly stochastic or mean-field assumptions.

Further, the closure error between the the macroscopic model and the aggregated microscopic in real networks is notably larger for the logistic model without the mean-field approximation, by at least an order of magnitude (Figure 4), confirming the theoretical results. During the active growth phase ($X \in [0.1, 0.8]$), the Gompertz closure error is consistently smaller than the logistic error — by approximately a factor of 2 on the heterogeneous Scale-Free network and a factor of 30 on the more homogeneous Erdős-Rényi network. Both errors diverge near saturation ($X \rightarrow 1$) where the theoretical growth rate approaches zero. The persistent logistic closure error reflects the network-state correlations ($\text{Cov}_W(\mathbf{x}, \mathbf{x}) > 0$) inherent to transmission dynamics, while the Gompertz error arises from column-sum deviations that are self-correcting as nodes synchronize, as well as machine-precision inaccuracies in taking the natural logarithm of numbers close to 1. We omit the discrepancy analysis for the mean-field approximated networks where the Gompertz closure is exact.

Microscopic Richards interpolation: θ -power-mean coupling, Log-Normal network, $N = 10,000$, $\beta = 0.25$, $\sigma = 0.02$

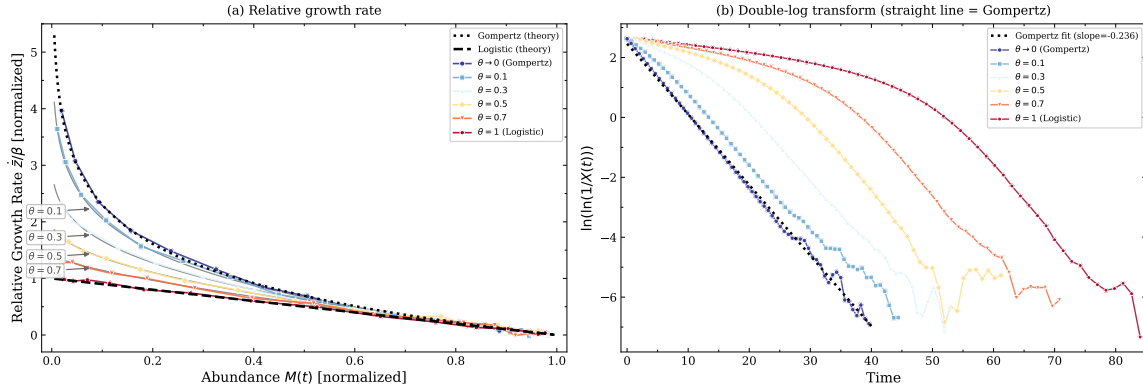


Figure 1: Different simulated Richard growth models are compared with their theoretical curves, using a Log-Normal network with 10,000 nodes and mean-field approximation. (a) Relative growth plotted against abundance and (b) Double-log transform $\ln(\ln(1/X(t)))$ versus time, where a straight line indicates Gompertz growth. Note that the higher values of θ makes the growth unstable due to higher mean-variance coupling as discussed in the text.

Microscopic Gompertz Growth: $\frac{dz_i}{dt} = -\beta z_i + \beta \sum_j W_{ij}(z_j - z_i)$, $N = 10,000$, $\beta = 0.25$, $\sigma = 0.001$

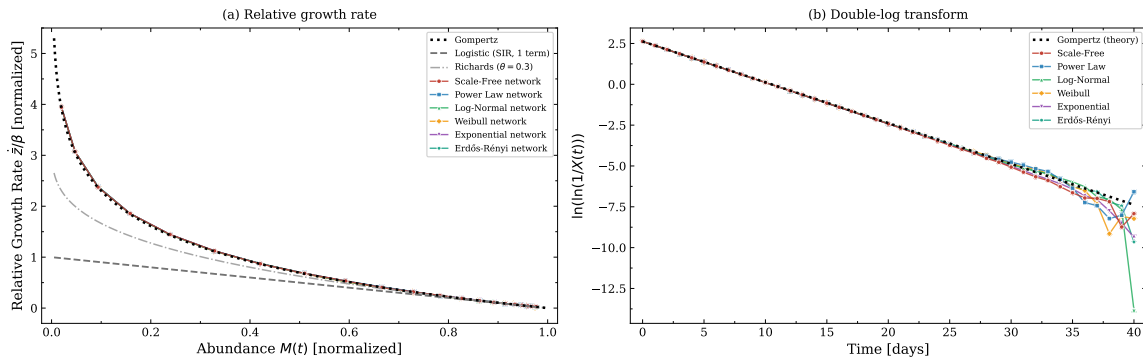


Figure 2: Comparison of relative growth rates for different networks under a microscopic Gompertz mode with 10,000 nodes using a mean field approximation, showing fidelity to the theoretical limit regardless of the underlying network. Different networks plotted in color, showing minimal deviation from the theoretical Gompertz curve. A logistic and Richards $\theta = 0.3$ theoretical curve is plotted for comparison. (a) Relative growth plotted against abundance and (b) Double-log transform $\ln(\ln(1/X(t)))$ versus time, where a straight line indicates Gompertz growth. The abundance variable is normalized to values between 0 and 1.

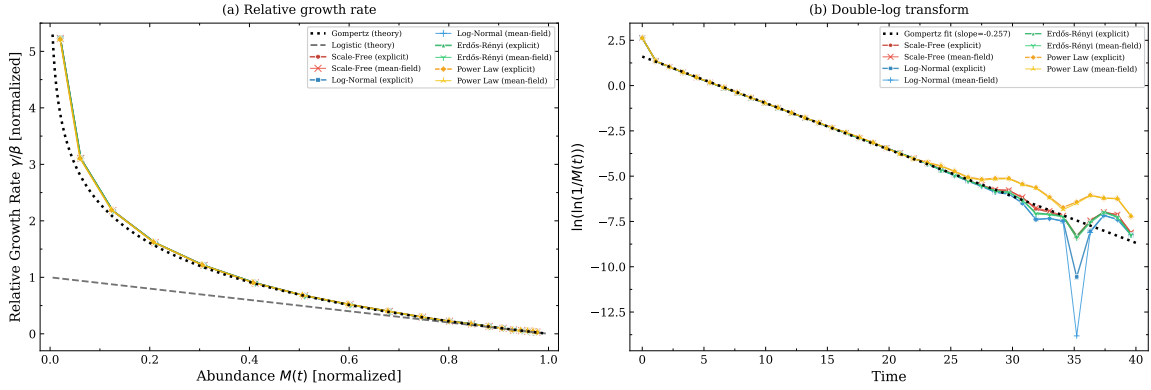
Explicit network vs mean-field: Gompertz growth ($\theta = 0$), $N = 10,000$, $\beta = 0.25$, $\sigma = 0.001$ 

Figure 3: Comparison of Gompertz growth ($\theta = 0$) on explicit configuration-model networks versus the mean-field approximation, for four network topologies: Scale-Free ($P(k) \propto k^{-3}$), Log-Normal, Erdős-Rényi, and Power Law ($P(k) \propto k^{-2.2}$). Each network has $N = 10,000$ nodes with row-normalized coupling $w_{ij} = a_{ij}/k_i$, where a_{ij} is the binary adjacency matrix and $k_i = \sum_j a_{ij}$ is the degree of node i . This normalization is not doubly stochastic for heterogeneous degree distributions. The explicit simulations compute each node's neighbor average from its actual neighbors in the sparse adjacency structure, with no global averaging. (a) Relative growth rate γ/β versus normalized abundance $M(t)$. (b) Double-log transform $\ln(\ln(1/X(t)))$ versus time, where a straight line indicates Gompertz growth. Parameters: $\beta = 0.25$, $\sigma = 0.001$, $\Delta t = 0.05$.

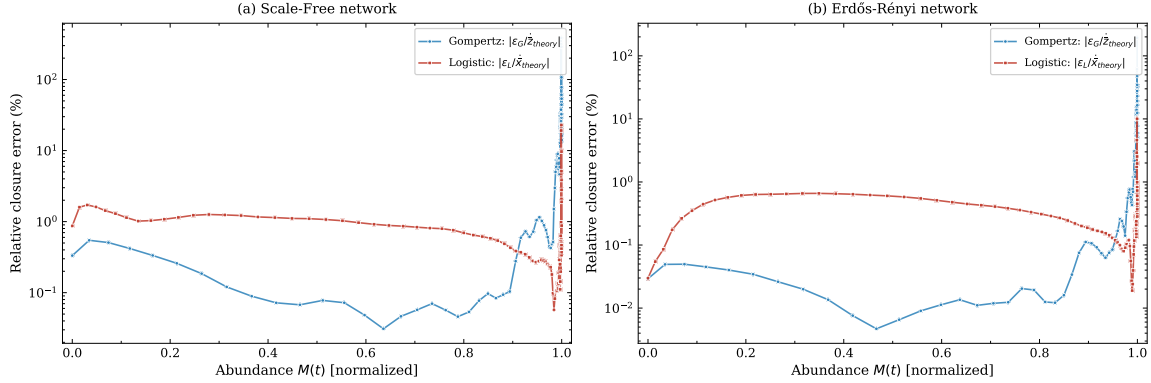
Relative and absolute macroscopic closure error on explicit networks (no mean-field)
 $N = 10,000$, $\beta = 0.25$, $\sigma = 0.0005$ 

Figure 4: Relative macroscopic closure error for Gompertz and logistic growth on explicit (non-mean-field) networks with row-normalized coupling $w_{ij} = a_{ij}/k_i$, where a_{ij} is the binary adjacency matrix and $k_i = \sum_j a_{ij}$ is the degree of node i . The Gompertz closure error is defined as $|\epsilon_G/\dot{z}_{\text{theory}}|$ where $\epsilon_G = \dot{z} - (-\beta\bar{z})$ measures the deviation of the actual macroscopic dynamics from the closed-form Gompertz equation. The logistic closure error is $|\epsilon_L/\dot{x}_{\text{theory}}|$ where $\epsilon_L = \dot{x} - \beta\bar{x}(1 - \bar{x})$ measures the deviation from the closed-form logistic equation. Both are computed on configuration-model random graphs with $N = 10,000$ nodes. (a) Scale-Free network ($P(k) \propto k^{-3}$, max degree 417). (b) Erdős-Rényi network ($k \sim \text{Poisson}(\lambda = 6)$, max degree 19). Parameters: $\beta = 0.25$, $\sigma = 0.0005$, $\Delta t = 0.02$.

5 Discussion and Conclusion

In conclusion, we have shown that Gompertzian growth is a maximally coherent growth process with specific mathematical properties: 1. closure under aggregation, 2. stability in spite of internal fluctuation, 3. mean-variance uncoupling, and 4. egalitarian forcing with equal contributions to the relative growth. All of these properties shed new light on dynamics exhibited at the macroscopic level.

We can now further appreciate the Richards parametrization as a transition from non-collaborative logistic growth to maximally collaborative Gompertz growth as $\theta \rightarrow 0$. This feature of θ was also obtained by Petroni et al. (2020) [19] by interpreting the θ -logistic growth rate as non-linear resource availability dependent on the overall abundance, with linearity reached at $\theta = 1$, without resorting to microscopic modelling. Coherence was there defined as maximally synergistic or collaborating elementary components. Used in a mortality model, the maximally coherent scenario implies that all units in the network are exposed to some external stressor simultaneously. Molski and Konarski (2003) [31] made a similar observation that Gompertzian growth is the coherent state in a quantum mechanical system with a time-dependent potential, an interpretation which sheds further light on the temporal nature of the postulated stress response. This quantum mechanical system has also been used to describe coherent energy states of diatomic molecules in space [69]. There is also an analogy between the mean-variance decoupling of the Gompertz at $\theta \rightarrow 0$ and symmetry-breaking dynamics in various physical systems. The spectral gap of the deviation dynamics $\dot{\delta} = \beta(-2I + W)\delta$ protect the macroscopic state from microscopic perturbations through a structural property of the dynamics. In the field of quantum physics, coherence is a well-defined mathematical property first explored by Glauber (1963) [32] in the context of electromagnetic fields. The fact that stable macroscopic dynamics emerge from the microscopic scales only from the Gompertz model compared to other Richards and logistic models is also noteworthy.

Growth of infinitely interacting systems where entities are classical in nature with well-defined position and momentum suggests the source of growth is a ubiquitous field, externally or internally created, which affects all microscopic entities instantaneously². With such a source, entities need not be interacting with each other but rather respond to the field-induced stimuli. Examples of such models are already given by De Lauro (2014), Sacher (1956) and Markov (2019). De Lauro et al. [29] and Sacher [28] consider the source of growth to be stochastic or random in nature, and while Lauro et al. do not discuss the external nature of such a source, their model does not preclude it. More generally, with growth as stochastically rooted the source is also ubiquitously present uniformly across all entities from the very onset of growth. Markov (2019)[30] discusses growth within the context of chemical reactions, where the source of growth is a catalytic agent that is implicitly available ubiquitously to all reactants to the same degree, exactly in line with De Lauro and Sacher’s models.

For many growth processes within biological organisms, it has been well established that the mediator can be the electromagnetic field [33], which suggests that the accompanying growth pattern in these processes follows the Gompertz model as the result of a ubiquitous growth-inducing field. For growth in larger ecological or societal systems other catalysts can be air pollution, oxygen deprivation, etc.

We finally note that the original formulation of the Gompertz model made by Gompertz himself was concerning mortality rates as a function of age. He noticed that mortality rates increased at a geometric rate with age. In this context the ubiquitous stressor postulated above would be the living conditions on our planet, for which time acts as a proxy. It is indeed curious that we do not see mortality rates behave as linearly dependent on time, for which a Gaussian curve would be appropriate. Nor do we see any finite-ordered polynomial time dependence on mortality rates. Instead we see growth of mortality rates as induced by a stressor, in this case time.

6 Methods

6.1 Running growth model simulations on networks

The simulations of growth in a theoretical network structure are done using Equation 12 with and without a mean-field approximation, using the shorthand notation $\dot{z} = \frac{1}{N} \sum_i \frac{dz_i}{dt}$. In the mean-field case, each node’s local neighborhood average is replaced by a degree-weighted global average. Specifically, the coupling target $\bar{z}_{\theta,i} = \frac{1}{\theta} \ln(\sum_j w_{ij} e^{\theta z_j})$ is approximated as $\bar{z}_{\theta} = \frac{1}{\theta} \ln(\sum_j w_j e^{\theta z_j})$ with weights $w_j = k_j / \sum_k k_k$, where $k_j = \sum_i a_{ij}$ is the degree of node j and a_{ij} is the binary adjacency matrix. This approximation is standard for configuration-model random graphs and becomes exact for large N when the correlation between node state z_j and degree k_j is weak.

In the explicit network case, no such approximation is made. A configuration-model random graph is constructed from the degree sequence by creating a stub list (each node i contributing k_i stubs), randomly pairing stubs to form edges, and discarding self-loops. The resulting adjacency structure is stored as a sparse neighbor list. The coupling weights are row-normalized as $w_{ij} = a_{ij}/k_i$, so that each node averages over its actual neighbors. This normalization is not doubly stochastic for heterogeneous degree distributions, meaning the column sums $c_j = \sum_i w_{ij} = \sum_{i \in \mathcal{N}(j)} 1/k_i$ vary across nodes. As shown in the text, this does not materially affect the macroscopic Gompertz dynamics because the column-sum deviations are self-correcting under the stable coupling.

²Or at speeds much greater than the differential times in the growth equations

We simulate $N = 10,000$ nodes with parameters $\beta = 0.25$, $\sigma = 0.001$, and time step $\Delta t = 0.02$. Initial conditions are drawn as $z_i \sim \mathcal{N}(-13.8, 0.03^2)$, corresponding to initial abundance $x_i \approx 10^{-6}$. Integration is performed using the Euler-Maruyama scheme. For θ near 1, the mean-variance coupling identified in Eq. 19 produces visible noise in the later stages of the simulations.

The node degrees are drawn from the following distributions, each representing a distinct network topology:

- **Scale-free:** $P(k) \propto k^{-3}$, $k_{\min} = 2$, generated via inverse CDF sampling, approximating a Barabási-Albert network.
- **Power law:** $P(k) \propto k^{-2.2}$, $k_{\min} = 2$, a heavier-tailed variant generated via inverse CDF sampling.
- **Log-normal:** $\ln k \sim \mathcal{N}(1.5, 0.8^2)$, $k_{\min} = 2$.
- **Weibull:** $k \sim \text{Weibull}(\text{shape} = 1.5, \text{scale} = 5)$, $k_{\min} = 2$.
- **Exponential:** $k \sim \text{Exp}(\text{scale} = 4)$, $k_{\min} = 2$.
- **Erdős-Rényi:** $k \sim \text{Poisson}(\lambda = 6)$, $k_{\min} = 2$.

For the Richards interpolation experiments, the coherence parameter θ is varied in $\{0, 0.1, 0.3, 0.5, 0.7, 1.0\}$ on a single network (log-normal degree distribution). Since logistic-like dynamics ($\theta \rightarrow 1$) exhibit slower growth from low initial abundance, the simulation duration is extended as $T_{\max} = 40 \cdot \max(1, 1 + 4\theta)$ to ensure all trajectories reach saturation. The macroscopic observable is the geometric mean $M(t) = \exp(\frac{1}{N} \sum_i z_i)$, and the relative growth rate is computed numerically as $\gamma(t) = \dot{M}(t)/M(t)$ using a central difference scheme for $\dot{M}$.

6.2 Measuring macroscopic closure error

To quantify the degree to which each model's macroscopic equation holds on real (non-mean-field) networks, we compute the relative closure error for both Gompertz and logistic dynamics. For the Gompertz model, the closure error is defined as

$$\epsilon_G(t) = \dot{\bar{z}}(t) - (-\beta \bar{z}(t)), \quad (30)$$

where $\dot{\bar{z}} = \frac{1}{N} \sum_i \dot{z}_i$ is the actual macroscopic rate of change computed from the microscopic dynamics, and $-\beta \bar{z}$ is the theoretical closed-form prediction. For the logistic model, the microscopic dynamics are $\dot{x}_i = \beta(1 - x_i) \sum_j w_{ij} x_j$, and the closure error is

$$\epsilon_L(t) = \dot{\bar{x}}(t) - \beta \bar{x}(t)(1 - \bar{x}(t)), \quad (31)$$

where $\dot{\bar{x}} = \frac{1}{N} \sum_i \dot{x}_i$ and $\beta \bar{x}(1 - \bar{x})$ is the macroscopic logistic prediction. The relative closure error is then $|\epsilon/\dot{X}_{\text{theory}}|$, which gives a dimensionless measure of how much the actual macroscopic dynamics deviate from the closed-form equation, normalized by the expected growth rate.

Both models are simulated on explicit configuration-model networks with row-normalized coupling $w_{ij} = a_{ij}/k_i$, using the same initial conditions ($z_i \sim \mathcal{N}(-2.3, 0.5^2)$, corresponding to $x_i \approx 0.1$) to ensure both models traverse the active growth phase during the simulation window. A smoothing window of 25 time steps is applied to the relative error for visual clarity. The logistic simulation uses multiplicative noise $\sqrt{\sigma} x_i dB_i$ consistent with the stochastic framework, while the Gompertz simulation uses additive noise $\sqrt{\sigma} dB_i$ in the log-domain, as specified in Eq. 12. Parameters: $N = 10,000$, $\beta = 0.25$, $\sigma = 0.0005$, $\Delta t = 0.02$, $T_{\max} = 50$.

Competing interests

The authors declare no competing interests.

Code availability

Code is available at [github.com/xyz]

References

- [1] Gompertz, B. XXIV. On the nature of the function expressive of the law of human mortality, and on a new mode of determining the value of life contingencies. in a letter to francis baily, esq. f. r. s. &c. *Philosophical Transactions of the Royal Society of London*, 115:513–583, dec 1825. doi:10.1098/rstl.1825.0026.

- [2] Makeham, W. M. On the further development of gompertz's law. *Journal of the Institute of Actuaries*, 28(2): 152–159, 1889.
- [3] Weymouth, F. W. and McMillin, H. C. *Relative growth and mortality of the Pacific razor clam (*Siliqua patula*, Dixon) and their bearing on the commercial fishery*. Number 1099. US Government Printing Office, 1931.
- [4] Weymouth, F. W. and Thompson, S. H. *The age and growth of the pacific cockle (*Cardium corbis*, Martyn)*. Number 1101. US Government Printing Office, 1931.
- [5] Laird, A. K. Dynamics of tumour growth. *British journal of cancer*, 18(3):490, 1964.
- [6] Zwietering, M. H., Jongenburger, I., Rombouts, F. M., and Van't Riet, K. Modeling of the bacterial growth curve. *Applied and environmental microbiology*, 56(6):1875–1881, 1990.
- [7] Skinner, G. E., Larkin, J. W., and Rhodhamel, E. J. Mathematical modeling of microbial growth: a review. *Journal of food safety*, 14(3):175–217, 1994.
- [8] Starck, J. M. and Ricklefs, R. E. *Avian growth and development: evolution within the altricial-precocial spectrum*. Number 8. Oxford University Press on Demand, 1998.
- [9] Aggrey, S. Comparison of three nonlinear and spline regression models for describing chicken growth curves. *Poultry science*, 81(12):1782–1788, 2002.
- [10] Paine, C. T., Marthews, T. R., Vogt, D. R., Purves, D., Rees, M., Hector, A., and Turnbull, L. A. How to fit nonlinear plant growth models and calculate growth rates: an update for ecologists. *Methods in Ecology and Evolution*, 3(2):245–256, 2012.
- [11] Benzekry, S., Lamont, C., Beheshti, A., Tracz, A., Ebos, J. M., Hlatky, L., and Hahnfeldt, P. Classical mathematical models for description and prediction of experimental tumor growth. *PLoS computational biology*, 10(8):e1003800, 2014.
- [12] Halmi, M., Shukor, M., Johari, W., and Shukor, M. Evaluation of several mathematical models for fitting the growth of the algae *dunaliella tertiolecta*. *Asian Journal of Plant Biology*, 2(1):1–6, 2014.
- [13] Tjørve, K. M. and Tjørve, E. Shapes and functions of bird-growth models: how to characterise chick postnatal growth. *Zoology*, 113(6):326–333, 2010.
- [14] Winsor, C. P. The gompertz curve as a growth curve. *Proceedings of the National Academy of Sciences*, 18(1): 1–8, jan 1932. doi:10.1073/pnas.18.1.1.
- [15] Olshansky, S. J. and Carnes, B. A. Ever since gompertz. *Demography*, 34:1–15, 1997.
- [16] Prescott, R. B. Law of growth in forecasting demand. *Journal of the American Statistical Association*, 18(140): 471–479, 1922.
- [17] Peabody, L. E. Growth curves and railway traffic. *Journal of the American Statistical Association*, 19(148): 476–483, 1924.
- [18] Bajzer, Ž., Vuk-Pavlović, S., and Huzak, M. Mathematical modeling of tumor growth kinetics. In *A survey of models for tumor-immune system dynamics*, pages 89–133. Springer, 1997.
- [19] Petroni, N. C., De Martino, S., and De Siena, S. Logistic and θ -logistic models in population dynamics: General analysis and exact results. *Journal of Physics A: Mathematical and Theoretical*, 53(44):445005, 2020.
- [20] Harko, T., Lobo, F. S., and Mak, M. Exact analytical solutions of the susceptible-infected-recovered (sir) epidemic model and of the sir model with equal death and birth rates. *Applied Mathematics and Computation*, 236:184–194, 2014.
- [21] Kröger, M. and Schlickeiser, R. Analytical solution of the SIR-model for the temporal evolution of epidemics. part a: time-independent reproduction factor. *Journal of Physics A: Mathematical and Theoretical*, 53(50):505601, 2020.
- [22] Schlickeiser, R. and Kröger, M. Analytical solution of the SIR-model for the temporal evolution of epidemics: part b. semi-time case. *Journal of Physics A: Mathematical and Theoretical*, 54(17):175601, 2021.
- [23] Heng, K. and Althaus, C. L. The approximately universal shapes of epidemic curves in the susceptible-exposed-infectious-recovered (SEIR) model. *Scientific Reports*, 10(1):1–6, 2020.
- [24] Berryman, A. A. The origins and evolution of predator-prey theory. *Ecology*, 73(5):1530–1535, 1992.
- [25] Neafsey, P. J., Boxenbaum, H., Ciraulo, D. A., and Fournier, D. J. A gompertz age-specific mortality rate model of aging, hormesis, and toxicity: Fixed-dose studies. *Drug metabolism reviews*, 19(3-4):369–401, 1988.
- [26] Neafsey, P. J., Ciraulo, D. A., Boxenbaum, H., and Fournier, D. J. A gompertz age-specific mortality rate model of aging, hormesis, and toxicity: dose-response studies. *Drug Metabolism Reviews*, 20(1):111–150, 1989.

- [27] Thompson, G. A., Smithers, J., and Boxenbaum, H. Biphasic mortality response of chipmunks in the wild to single doses of ionizing radiation: toxicity and longevity hormesis. *Drug metabolism reviews*, 22(2-3):269–289, 1990.
- [28] Sacher, G. A. On the statistical nature of mortality, with especial reference to chronic radiation mortality. *Radiology*, 67(2):250–258, 1956.
- [29] De Lauro, E., De Martino, S., De Siena, S., and Giorno, V. Stochastic roots of growth phenomena. *Physica A: Statistical Mechanics and its Applications*, 401:207–213, 2014.
- [30] Markov, S. M. Reaction networks reveal new links between gompertz and verhulst growth functions. *Biomath*, 8(1):ID–1904167, 2019.
- [31] Molski, M. and Konarski, J. Coherent states of gompertzian growth. *Physical review E*, 68(2):021916, 2003.
- [32] Glauber, R. J. Coherent and incoherent states of the radiation field. *Physical Review*, 131(6):2766, 1963.
- [33] Becker, R. O. Electromagnetic controls over biological growth processes. *Journal of Bioelectricity*, 3(1-2): 105–118, 1984.
- [34] Langman, L. and Burr, H. A technique to aid in the detection of malignancy of the female genital tract. *American Journal of Obstetrics & Gynecology*, 57(2):274–281, 1949.
- [35] Kozusko, F. and Bajzer, Ž. Combining gompertzian growth and cell population dynamics. *Mathematical biosciences*, 185(2):153–167, 2003.
- [36] Goodman, R., Bassett, C. A. L., and Henderson, A. S. Pulsing electromagnetic fields induce cellular transcription. *Science*, 220(4603):1283–1285, 1983.
- [37] Liboff, A., Williams Jr, T., Strong, D., and Wistar Jr, R. Alternating magnetic fields enhance dna synthesis in fibroblastic cells. *Bioelectrical Repair and Growth Society (BRAGS)*, 1981.
- [38] Norton, L., Hanley, K., and Turkewicz, J. Bioelectric perturbations of bone: Research directions and clinical applications. *The Angle Orthodontist*, 54(1):73–87, 1984.
- [39] Liboff, A., Williams Jr, T., Strong, D., and Wistar Jr, R. Time-varying magnetic fields: effect on dna synthesis. *Science*, 223(4638):818–820, 1984.
- [40] Kendal, W. Gompertzian growth as a consequence of tumor heterogeneity. *Mathematical biosciences*, 73(1): 103–107, 1985.
- [41] Wang, X.-S., Wu, J., and Yang, Y. Richards model revisited: Validation by and application to infection dynamics. *Journal of theoretical biology*, 313:12–19, 2012.
- [42] Kermack, W. O. and McKendrick, A. G. A contribution to the mathematical theory of epidemics. *Proceedings of the royal society of london. Series A, Containing papers of a mathematical and physical character*, 115(772): 700–721, 1927.
- [43] Richards, F. A flexible growth function for empirical use. *Journal of experimental Botany*, 10(2):290–301, 1959.
- [44] Vardavas, R., de Lima, P. N., Davis, P. K., Parker, A. M., and Baker, L. Modeling infectious behaviors: the need to account for behavioral adaptation in covid-19 models. *Policy and complex systems*, 7(1):21, 2021.
- [45] Osi, A. and Ghaffarzadegan, N. Parameter estimation in behavioral epidemic models with endogenous societal risk-response. *PLOS Computational Biology*, 20(3):e1011992, 2024.
- [46] Lloyd-Smith, J. O., Schreiber, S. J., Kopp, P. E., and Getz, W. M. Superspreading and the effect of individual variation on disease emergence. *Nature*, 438(7066):355–359, 2005.
- [47] Zonta, F. and Levitt, M. Virus spread on a scale-free network reproduces the gompertz growth observed in isolated covid-19 outbreaks. *Advances in Biological Regulation*, 86:100915, 2022.
- [48] Clauset, A., Shalizi, C. R., and Newman, M. E. Power-law distributions in empirical data. *SIAM review*, 51(4): 661–703, 2009.
- [49] Broido, A. D. and Clauset, A. Scale-free networks are rare. *Nature communications*, 10(1):1–10, 2019.
- [50] Holme, P. Rare and everywhere: Perspectives on scale-free networks. *Nature communications*, 10(1):1016, 2019.
- [51] Herrmann, H. A. and Schwartz, J.-M. Why COVID-19 models should incorporate the network of social interactions. *Physical Biology*, 17(6):065008, 2020.
- [52] Carletti, T., Fanelli, D., and Piazza, F. COVID-19: The unreasonable effectiveness of simple models. *Chaos, Solitons & Fractals: X*, 5:100034, 2020.

- [53] Cooper, I., Mondal, A., and Antonopoulos, C. G. A SIR model assumption for the spread of COVID-19 in different communities. *Chaos, Solitons & Fractals*, 139:110057, 2020.
- [54] Postnikov, E. B. Estimation of COVID-19 dynamics “on a back-of-envelope”: Does the simplest SIR model provide quantitative parameters and predictions? *Chaos, Solitons & Fractals*, 135:109841, 2020.
- [55] Muñoz-Fernández, G. A., Seoane, J. M., and Seoane-Sepúlveda, J. B. A SIR-type model describing the successive waves of COVID-19. *Chaos, Solitons & Fractals*, 144:110682, 2021.
- [56] Cooper, I., Mondal, A., Antonopoulos, C. G., and Mishra, A. Dynamical analysis of the infection status in diverse communities due to COVID-19 using a modified SIR model. *Nonlinear Dynamics*, pages 1–14, 2022.
- [57] Saikia, D., Bora, K., and Bora, M. P. COVID-19 outbreak in india: an SEIR model-based analysis. *Nonlinear Dynamics*, 104(4):4727–4751, 2021.
- [58] Ohnishi, A., Namekawa, Y., and Fukui, T. Universality in COVID-19 spread in view of the gompertz function. *Progress of Theoretical and Experimental Physics*, 2020(12), oct 2020. doi:10.1093/ptep/ptaa148.
- [59] Rypdal, K. and Rypdal, M. A parsimonious description and cross-country analysis of COVID-19 epidemic curves. *International Journal of Environmental Research and Public Health*, 17(18):6487, sep 2020. doi:10.3390/ijerph17186487.
- [60] Català, M., Alonso, S., Alvarez-Lacalle, E., López, D., Cardona, P.-J., and Prats, C. Empirical model for short-time prediction of COVID-19 spreading. *PLOS Computational Biology*, 16(12):e1008431, dec 2020. doi:10.1371/journal.pcbi.1008431.
- [61] Rodrigues, T. and Helene, O. Monte carlo approach to model COVID-19 deaths and infections using gompertz functions. *Physical Review Research*, 2(4):043381, 2020.
- [62] Levitt, M., Scaiewicz, A., and Zonta, F. Predicting the trajectory of any COVID19 epidemic from the best straight line. *medRxiv*, jun 2020. doi:10.1101/2020.06.26.20140814.
- [63] Tjørve, K. M. and Tjørve, E. The use of gompertz models in growth analyses, and new gompertz-model approach: An addition to the unified-richards family. *PloS one*, 12(6):e0178691, 2017.
- [64] Sinkhorn, R. and Knopp, P. Concerning nonnegative matrices and doubly stochastic matrices. *Pacific Journal of Mathematics*, 21(2):343–348, 1967.
- [65] Zhang, C.-L. and Popp, F.-A. Log-normal distribution of physiological parameters and the coherence of biological systems. *Medical Hypotheses*, 43(1):11–16, 1994.
- [66] Petroni, N. C., De Martino, S., and De Siena, S. Gompertz and logistic stochastic dynamics: Advances in an ongoing quest. *arXiv preprint arXiv:2002.06409*, 2020.
- [67] Gauss, C. F. and Davis, C. H. *Theory of the motion of the heavenly bodies moving about the sun in conic sections: a translation of Gauss’s “Theoria Motus.” with an appendix*. Little, Brown, 1857.
- [68] Stahl, S. The evolution of the normal distribution. *Mathematics magazine*, 79(2):96–113, 2006.
- [69] Morse, P. M. Diatomic molecules according to the wave mechanics. ii. vibrational levels. *Physical review*, 34(1): 57, 1929.